

RAJALAKSHMI INSTITUTE OF TECHNOLOGY

(An Autonomous Institution, Affiliated to Anna University, Chennai)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

ACADEMIC YEAR 2025 - 2026

SEMESTER IV

AD23421 – DATA ANALYTICS LABORATORY

MINI PROJECT REPORT

REGISTER NUMBER	**2117240070193**
NAME	**MUKESH P**
PROJECT TITLE	**Weather Data Analytics Project**
DATE OF SUBMISSION	
FACULTY IN-CHARGE	**Dr. T.VINCENT GNANARAJ**

Signature of Faculty In-charge **Signature of HoD**

Internal Examiner **External Examiner**

TABLE OF CONTENTS

CHAPTER NO.	TITLE		PAGE NO.
1.	**INTRODUCTION**		1
2.	**PROBLEM STATEMENT**		2
3.	**GOAL**		3
4.	**THEORETICAL BACKGROUND**		4
	4.1.	Data Analytics	4
	4.2	Weather Data	4
	4.3	Data Preprocessing	4
	4.4	Statistical Analysis Techniques	5
	4.5	Correlation Analysis	5
	4.6	Data Visualization	6
	4.7	Literature Survey	6
	4.8	Justification for Choosing the Algorithm	6
5.	**METHODOLOGY**		7
	5.1	Data Collection	7
	5.2	Data Cleaning and Preprocessing	7
	5.3	Data Transformation	8
	5.4	Exploratory Data Analysis (EDA)	8
	5.5	Statistical and Correlation Analysis	8
	5.6	Data Visualization	8
	5.7	Insight Generation	9
6.	**IMPLEMENTATION CODE AND DATASET USED**		10
	6.1	Dataset Used	10

	6.2	Python code	10
7.	**OUTPUT**		13
8.	**RESULTS AND FUTURE ENHANCEMENT**		16
	8.1	Temperature Analysis	16
	8.2	Rainfall Analysis	16
	8.3	Humidity Analysis	16
	8.4	Humidity vs Rainfall Relationship	16
	8.5	AQI Analysis	16
	8.6	Wind Speed vs Temperature	16
	8.7	Pressure vs Cloud Cover	17
	8.8	Correlation Analysis	17
	8.9	Overall Observation	17
9.	**REFERENCES**		18

LIST OF FIGURES

FIGURE NO	NAME OF THE FIGURE	PAGE NO
1	Correlation Heatmap	13
2	AQI Distribution	13
3	Temperature Across Cities	14
4	Wind Speed vs Temperature	14
5	Humidity vs Rainfall	14
6	Rainfall comparison	15
7	Pressure vs Cloud Cover	15

LIST OF TABLES

TABLE NO	NAME OF THE TABLE	PAGE NO
1	Sample Dataset	9
2	Dataset Description	10

1. INTRODUCTION

Data analytics is the process of collecting, organizing, and analyzing data to discover meaningful patterns, relationships, and insights. It plays a crucial role in modern decision-making by transforming raw data into useful information. Instead of relying on assumptions or guesswork, data analytics enables accurate conclusions based on real data, thereby improving the quality of decisions in various fields.

In this project, data analytics is applied to a weather dataset that includes important climatic parameters such as temperature, humidity, rainfall, wind speed, and Air Quality Index (AQI). These factors are essential in understanding environmental conditions and how they vary across different locations. By analyzing these variables, the project aims to explore how different weather elements are interconnected and how they influence each other.

The analysis involves key steps such as data cleaning, statistical analysis, and data visualization. Data cleaning ensures that the dataset is free from errors, missing values, and inconsistencies. Statistical techniques are used to identify trends and relationships, while visualization methods such as charts and graphs help in presenting the data in a clear and understandable manner. These techniques together make it easier to interpret complex data.

Analyzing weather data is highly significant as climate conditions directly impact various aspects of daily life, including agriculture, transportation, public health, and environmental planning. Understanding patterns such as temperature variations, rainfall distribution, and air quality changes can help in better planning and predicting future environmental conditions.

In addition, this project emphasizes the importance of identifying correlations between different weather parameters. For example, analyzing the relationship between temperature and humidity or rainfall and air quality can provide deeper insights into climate behavior. Such relationships are useful in predicting environmental changes and understanding seasonal variations across different regions.

Furthermore, the project highlights the role of data visualization in simplifying complex datasets. Graphs, charts, and visual representations make it easier to identify trends and communicate findings effectively. Clear visualization not only improves understanding but also helps in presenting the results to users, researchers, or decision-makers in a more meaningful way.

The main objective of this project is to analyze the given weather dataset to identify patterns, trends, and relationships among different climatic variables. The project also focuses on presenting the results using effective visualizations, making the insights easy to understand and useful for practical applications.

2. PROBLEM STATEMENT

Weather data consists of multiple interrelated variables such as temperature, humidity, rainfall, wind speed, atmospheric pressure, cloud cover, and air quality index (AQI). These factors collectively influence climatic conditions and environmental behavior. However, due to the complexity and volume of such data, it becomes difficult to manually analyze and understand the relationships between these variables. Without proper analysis, identifying patterns, trends, and meaningful insights from weather data is challenging.

In the given dataset, weather information is collected across multiple cities, capturing different climatic conditions at a specific point in time. Each city exhibits unique characteristics influenced by geographical location, environmental factors, and atmospheric conditions. As a result, variations in temperature, rainfall, humidity, and air quality are observed across different regions. Understanding these variations and how the variables interact with each other is essential for gaining insights into climate behavior.

The main problem addressed in this project is to analyze and interpret the weather dataset to identify patterns, relationships, and variations among different climatic variables. This involves cleaning the data, handling inconsistencies, performing statistical analysis, and using visualization techniques to represent the data in a meaningful way. One of the key challenges is determining whether strong relationships exist between variables such as humidity and rainfall, temperature and wind speed, or pressure and cloud cover.

Additionally, the project aims to examine the distribution of air quality index (AQI) values to understand environmental conditions across cities. Since air quality has a direct impact on human health and environmental sustainability, analyzing AQI patterns is an important aspect of the study.

By applying data analytics techniques, this project seeks to transform raw weather data into useful insights that are easy to interpret. The findings can help in understanding climatic behavior, identifying independent and dependent variables, and highlighting environmental patterns. Overall, the problem focuses on extracting meaningful information from complex weather data and presenting it in a clear, structured, and visually understandable format.

Furthermore, the project also addresses the challenge of making the analyzed data actionable and meaningful for real-world applications. Simply identifying patterns is not sufficient unless the results can be interpreted effectively for practical use. Therefore, this study emphasizes converting analytical results into clear insights that can support better understanding of environmental conditions, assist in future planning, and contribute to informed decision-making related to climate and sustainability.

3. GOAL

The goal of this project is to analyze the given weather dataset and extract meaningful insights about different climatic variables such as temperature, humidity, rainfall, wind speed, and air quality index (AQI). The expected result is to identify patterns, trends, and relationships between these variables using statistical analysis and data visualization techniques.

Through this analysis, the project aims to:

• Understand how weather conditions vary across different cities

• Identify whether any strong relationships exist between variables like humidity and rainfall or temperature and wind speed

• Present the findings in a clear and visual manner for easy interpretation

The possible outcomes of this project include gaining insights into climate behavior, recognizing the independence or dependence between weather variables, and highlighting environmental patterns such as variations in air quality. These results can be useful for basic climate understanding and can be extended further for applications like weather prediction, environmental monitoring, and decision-making in areas such as agriculture and urban planning.

In addition, the project aims to develop a systematic approach to handling real-world datasets by applying key data analytics techniques such as data preprocessing, statistical evaluation, and visualization. This helps in improving the ability to work with structured data, identify inconsistencies, and ensure accurate analysis, which are essential skills in the field of data science.

Furthermore, the project focuses on enhancing the interpretation of analytical results by converting numerical data into meaningful insights. By effectively using graphs and charts, the study ensures that complex information is presented in a simple and understandable form, making it easier for users, researchers, and decision-makers to draw conclusions and apply the findings in practical scenarios.

Moreover, this project also aims to build a foundational understanding of how data-driven approaches can be applied to solve real-world environmental problems. By analyzing weather data systematically, the study encourages logical thinking, pattern recognition, and analytical reasoning, which are crucial for developing effective solutions in the fields of data analytics and environmental science.

4. THEORETICAL BACKGROUND

Weather Data Analysis is an important application of Data Analytics that focuses on understanding climatic patterns and relationships between environmental variables.It combines concepts from data processing, statistical analysis, and visualization techniques to analyze weather conditions and generate meaningful insights.

4.1.Data Analytics

Data Analytics involves collecting, processing, and analyzing data to extract useful information. In the context of weather analysis, it helps in understanding climate behavior, identifying patterns, and studying relationships between variables such as temperature, humidity, rainfall, and wind speed.

Types of analytics involved:

- **Descriptive Analytics:** Summarizes weather data (e.g., average temperature, rainfall distribution)
- **Diagnostic Analytics**: Identifies relationships between variables (e.g., humidity vs rainfall)
- **Predictive Analytics**: Can be used to forecast future weather trends

4.2.Weather Data

Weather data is structured and consists of multiple environmental parameters such as:

- Temperature (Maximum, Minimum, Average)
- Humidity
- Rainfall
- Wind Speed
- Air Quality Index (AQI)
- Pressure and Cloud Cover

This data helps in analyzing climatic conditions across different cities and identifying variations in weather patterns.

4.3.Data Preprocessing

Data preprocessing is necessary to prepare raw data for analysis. It improves data quality and ensures accurate results.

Steps involved:

- Handling missing values
- Removing unwanted spaces or inconsistencies in column names
- Converting data into proper formats (e.g., date format)
- Ensuring uniform structure of the dataset

These steps make the dataset clean and ready for analysis.

4.4.Statistical Analysis Techniques

Statistical methods are used to summarize and analyze weather data.

- **Mean (Average):**

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$

- **Standard Deviation:**

$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2}$$

These measures help in understanding the distribution and variation of weather variables.

4.5.Correlation Analysis

Correlation analysis is used to measure the relationship between two variables.

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

Where:

- r= correlation coefficient
- x_i, y_i= individual values
- $\bar{x}, \bar{y}$ = mean values

This is used to analyze relationships such as:

- Humidity vs Rainfall

- Temperature vs Wind Speed

4.6.Data Visualization

Visualization helps in presenting weather data in a simple and understandable way. Common techniques include:

- Line charts (temperature trends)
- Bar charts (rainfall comparison across cities)
- Scatter plots (relationship between variables)
- Heatmaps (correlation between features)

These visual tools improve interpretation and make patterns easier to understand.

4.7.Literature Survey

Weather data analysis has been widely studied using various techniques:

- Traditional statistical methods are used to analyze climate patterns and seasonal variations
- Machine learning algorithms such as Linear Regression and Decision Trees are used for weather prediction
- Time series models like ARIMA are applied for forecasting weather trends

These approaches show that analyzing weather data is important for understanding climate behavior and improving predictions.

4.8.Justification for Choosing the Algorithm

Descriptive statistics and correlation analysis are chosen because:

- They are simple and easy to understand
- They provide clear insights into data patterns
- They require less computational complexity
- They are suitable for analyzing relationships without prediction

Since the main objective of this project is to understand and analyze weather data rather than build complex prediction models, these techniques are appropriate and effective.

5. METHODOLOGY

The methodology of this project follows a structured data analytics pipeline to process, analyze, and interpret weather data. The workflow consists of multiple stages, starting from data collection to final insight generation. Each step ensures accurate analysis and meaningful results.

5.1. Data Collection

The first step involves collecting relevant weather data for analysis. The dataset is obtained from publicly available sources such as Kaggle.

The dataset includes weather-related attributes such as:

- Date
- City
- State
- Temperature (Maximum, Minimum, Average)
- Humidity
- Rainfall
- Wind Speed
- Air Quality Index (AQI)
- Pressure
- Cloud Cover

This data forms the foundation for analyzing climatic patterns across different cities.

5.2. Data Cleaning and Preprocessing

Raw weather data may contain inconsistencies such as missing values or formatting issues. Therefore, preprocessing is required to improve data quality.

The following steps are performed:

- Handling missing or null values
- Removing extra spaces in column names
- Converting data into proper formats (e.g., date format)
- Ensuring consistency in numerical values

These steps ensure that the dataset is clean and ready for analysis.

5.3. Data Transformation

After preprocessing, the data is transformed into a suitable format for analysis.
Key steps include:

- Converting date into structured format (if needed)
- Extracting useful features (e.g., month, year)
- Selecting relevant variables for analysis
- Organizing data for easier visualization

This step ensures that all data is properly structured for further processing.

5.4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis is performed to understand patterns and relationships within the dataset.

The following analyses are conducted:

- Distribution of temperature across cities
- Rainfall comparison between different regions
- Relationship between humidity and rainfall
- Analysis of AQI distribution across cities
- Relationship between wind speed and temperature

EDA helps in identifying patterns and understanding the behavior of weather variables.

5.5. Statistical and Correlation Analysis

Statistical techniques are applied to summarize the dataset and identify relationships.

Steps involved:

- Calculation of mean, standard deviation, and range
- Correlation analysis between variables
- Identification of strong and weak relationships

This step helps in understanding how different weather variables are connected.

5.6. Data Visualization

Visualization is used to present the analysis results in a clear and understandable format. The following visualizations are generated:

- Line charts: Temperature trends
- Bar charts: Rainfall comparison across cities
- Scatter plots: Relationship between variables
- Heatmaps: Correlation between weather features
- Histograms: Distribution of AQI

These visual tools help in better interpretation of the data.

5.7. Insight Generation

Based on the analysis and visualizations, meaningful insights are derived, such as:

- Variation of weather conditions across cities
- Weak or strong relationships between climatic variables
- Identification of patterns in rainfall and temperature
- Understanding air quality variations (AQI levels)

SAMPLE DATASET

Table 1: Sample Dataset

Date	City	State	Temperature_Max	Temperature_Min	Temperature_Avg	Humidity	Rainfall	Wind_Speed	AQI	AQI_Category	Pressure	Cloud_Cover
01-01-2024	Mumbai	Maharashtra	32.5	18	25.2	77.6	0	3.3	259	Poor	1020.3	62.1
01-01-2024	Delhi	Delhi	25.4	10.7	18.1	84.1	0	9	130	Moderate	1008.4	46
01-01-2024	Bengaluru	Karnataka	37.2	30.8	34	49	3.7	6.6	54	Satisfactory	1008	61.3
01-01-2024	Chennai	Tamil Nadu	37.2	30.4	33.8	34.2	9.5	9	176	Moderate	993.4	70
01-01-2024	Kolkata	West Bengal	27.4	17.5	22.5	32.2	9.1	9.2	97	Satisfactory	1008.2	56.9
01-01-2024	Hyderabac	Telangana	44.4	31.6	38	91.1	51.9	4	140	Moderate	996.9	9.3
01-01-2024	Ahmedaba	Gujarat	32.8	25.1	28.9	83.9	0	5.2	345	Very Poor	1018.1	12.1
01-01-2024	Jaipur	Rajasthan	40.4	33.5	37	30.4	8.2	19.7	75	Satisfactory	992.6	39.1
01-01-2024	Lucknow	Uttar Pradesh	42.3	31	36.6	51.5	0	18.8	186	Moderate	1012.3	89.3
01-01-2024	Bhopal	Madhya Prade	27.4	15.3	21.3	79.5	0	14	73	Satisfactory	1005	7.4
02-01-2024	Mumbai	Maharashtra	25.6	14.3	19.9	50.4	0	11.4	63	Satisfactory	1016.4	26.7
02-01-2024	Delhi	Delhi	30.8	24.2	27.5	90.4	54.3	20.5	207	Poor	996.5	89.8
02-01-2024	Bengaluru	Karnataka	41.1	27.2	34.2	50.7	0	20.8	198	Moderate	1020.1	5.7
02-01-2024	Chennai	Tamil Nadu	33.3	26.1	29.7	37.8	0	13.9	341	Very Poor	1014.6	39.5
02-01-2024	Kolkata	West Bengal	44.2	36.7	40.5	62.3	0	16	126	Moderate	1007.6	9.9
02-01-2024	Hyderabac	Telangana	43.2	35.8	39.5	39.4	0	17.5	265	Poor	1016.7	27.6
02-01-2024	Ahmedaba	Gujarat	32.4	21	26.7	71.2	5.4	9.4	223	Poor	996.5	8.9
02-01-2024	Jaipur	Rajasthan	38.6	33.4	36	63.3	0	17.9	82	Satisfactory	1003.5	94
02-01-2024	Lucknow	Uttar Pradesh	31.8	25.7	28.8	90.1	8.8	20.8	114	Moderate	1009.4	55.3

6. IMPLEMENTATION CODE AND DATASET USED

6.1. Dataset Used – https://www.kaggle.com/datasets/ankushnarwade/indian-climate-dataset- 20242025

Table 2: Dataset Description

Category	Description / Example
Dataset Name	Indian_Climate_Dataset_2024_2025
Domain	Weather
Objective	Descriptive Analysis and Relationship Identification
Data Source	Public
Number of Samples	7310
Data Type	Tabular
Instance Description	Each record represents weather data of a specific city on a particular date
Features (Inputs)	Temperature_Max, Temperature_Min, Temperature_Avg, Humidity, Rainfall, Wind_Speed, AQI, Pressure, Cloud_Cover
Feature Types	Numerical (most features), Categorical (City, State, AQI_Category)
License	Publicly Available
Version	Version 1.0

6.2.Python code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df =
pd.read_csv("DA\Weather_data_analytics\Indian_Climate_Dataset_2024_2025.csv")
df.columns = df.columns.str.strip()
print(df.columns)
# 1. BASIC CHECK
print(df.head())
```

```
print(df.info())

# 2. HANDLE MISSING VALUES
print("\nMissing values:\n", df.isnull().sum())

df.fillna(method='ffill', inplace=True)

# 3. DESCRIPTIVE STATS
print("\nStatistics:\n")
print(df.describe())

# 4. CORRELATION HEATMAP
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Between Weather Variables")
plt.show()

# 5. VISUALIZATIONS

# Temperature comparison
plt.figure(figsize=(8,5))
plt.plot(df['City'], df['Temperature_Max'], label='Max Temp')
plt.plot(df['City'], df['Temperature_Min'], label='Min Temp')
plt.plot(df['City'], df['Temperature_Avg'], label='Avg Temp')
plt.xticks(rotation=45)
plt.title("Temperature Comparison Across Cities")
plt.legend()
plt.show()

# Rainfall comparison
plt.figure(figsize=(8,5))
sns.barplot(x='City', y='Rainfall', data=df)
plt.xticks(rotation=45)
plt.title("Rainfall Across Cities")
plt.show()

# Humidity vs Rainfall
plt.figure(figsize=(8,5))
sns.scatterplot(x='Humidity', y='Rainfall', data=df)
plt.title("Humidity vs Rainfall")
plt.show()

# AQI distribution
plt.figure(figsize=(8,5))
```

```
sns.histplot(df['AQI'], bins=10, kde=True)
plt.title("AQI Distribution")
plt.show()

# Wind Speed vs Temperature
plt.figure(figsize=(8,5))
sns.scatterplot(x='Wind_Speed', y='Temperature_Avg', data=df)
plt.title("Wind Speed vs Average Temperature")
plt.show()

# Pressure vs Cloud Cover
plt.figure(figsize=(8,5))
sns.scatterplot(x='Pressure', y='Cloud_Cover', data=df)
plt.title("Pressure vs Cloud Cover")
plt.show()
```

7. OUTPUT

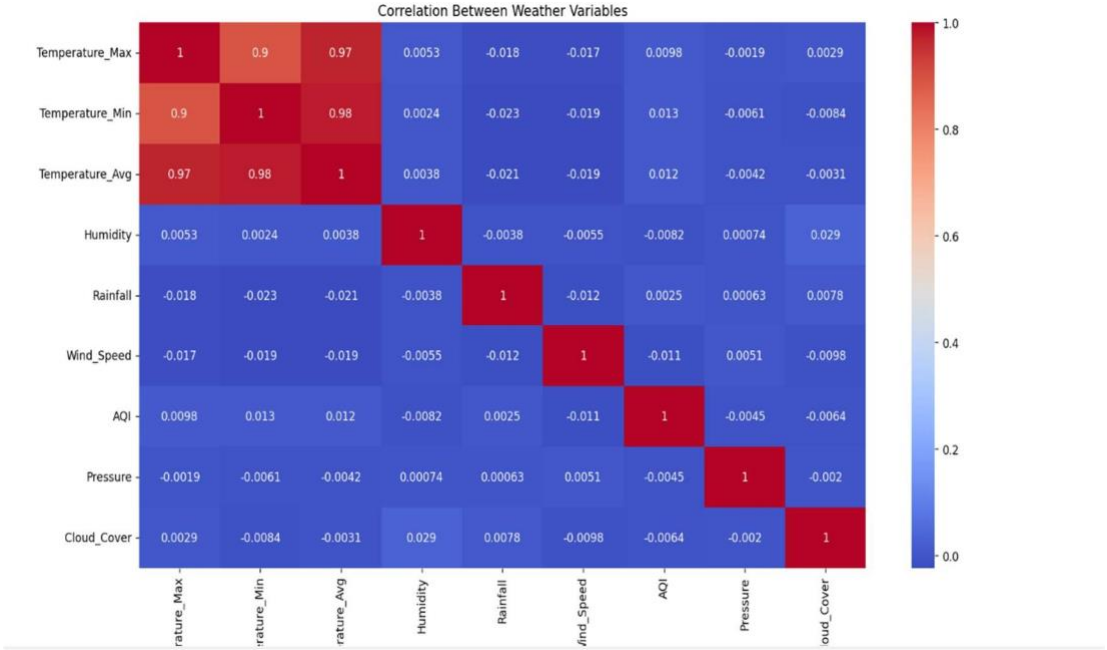


Figure 1: Correlation Heatmap

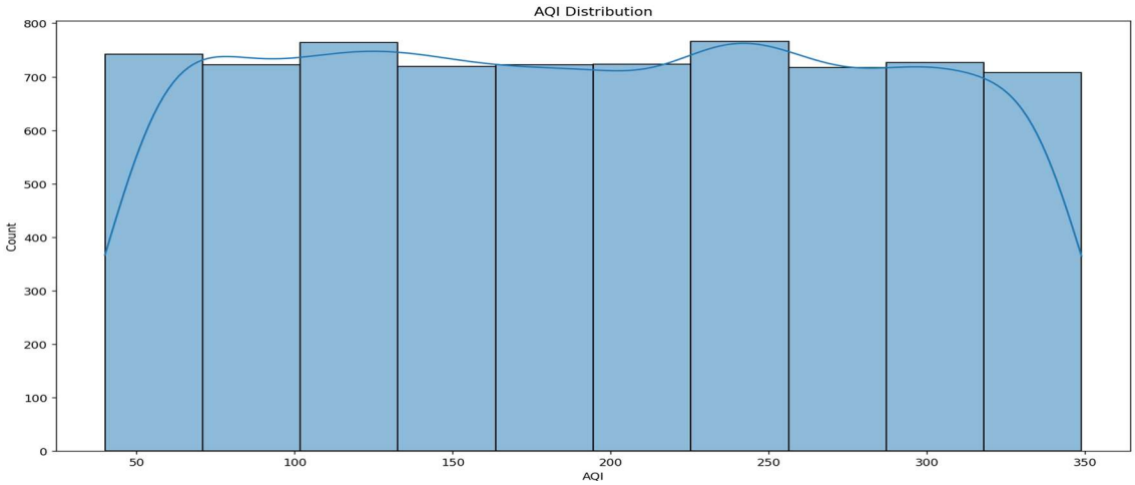


Figure 2: AQI Distribution

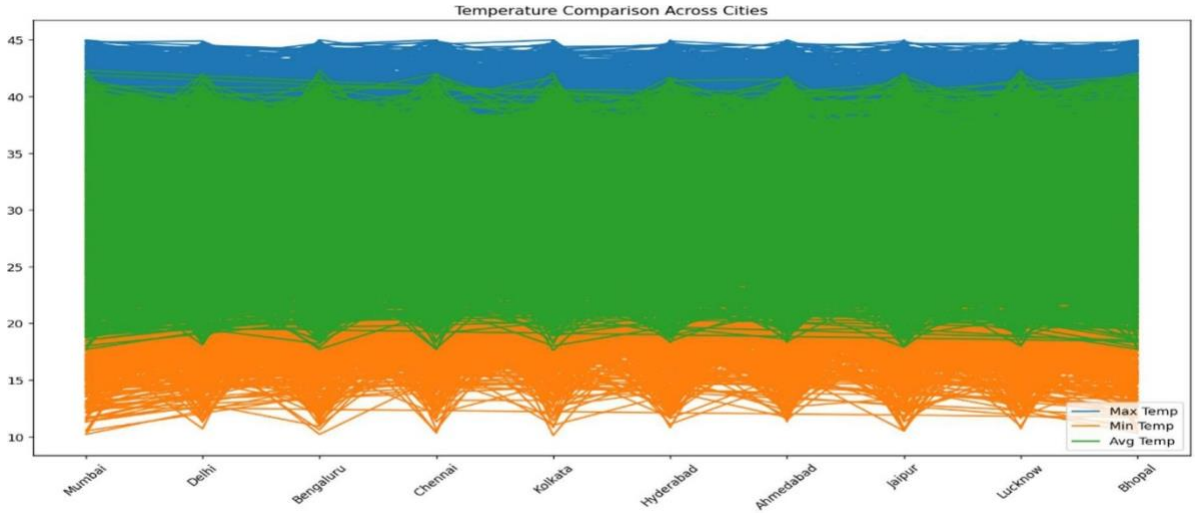


Figure 3: Temperature Across Cities

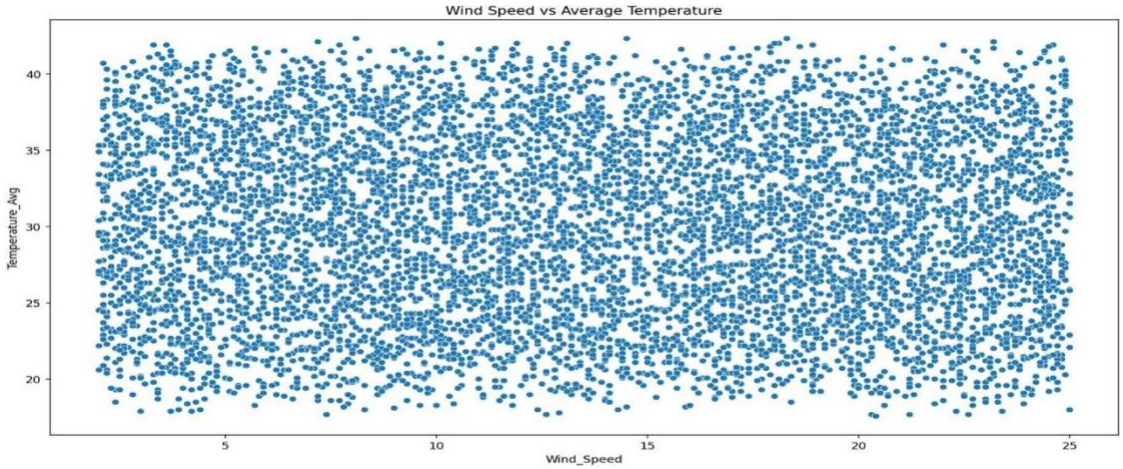


Figure 4: Wind Speed vs Temperature

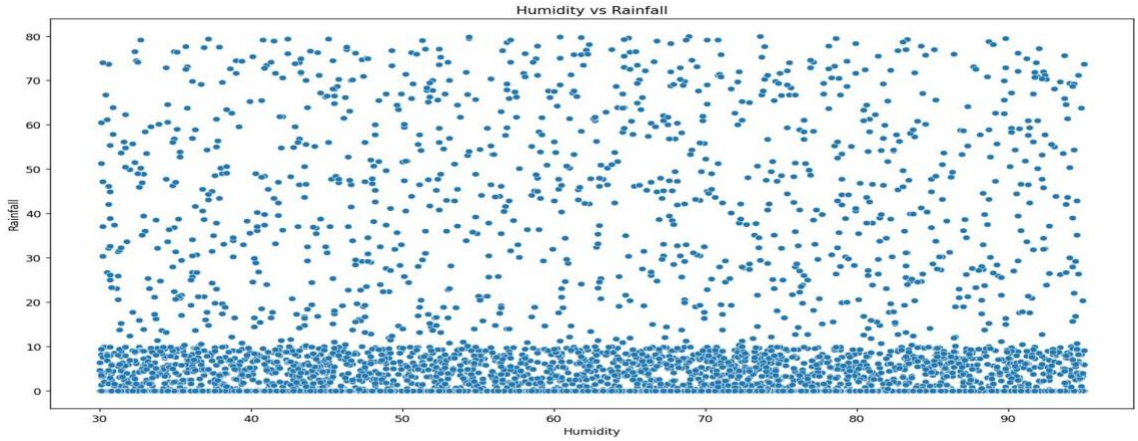


Figure 5: Humidity vs Rainfall

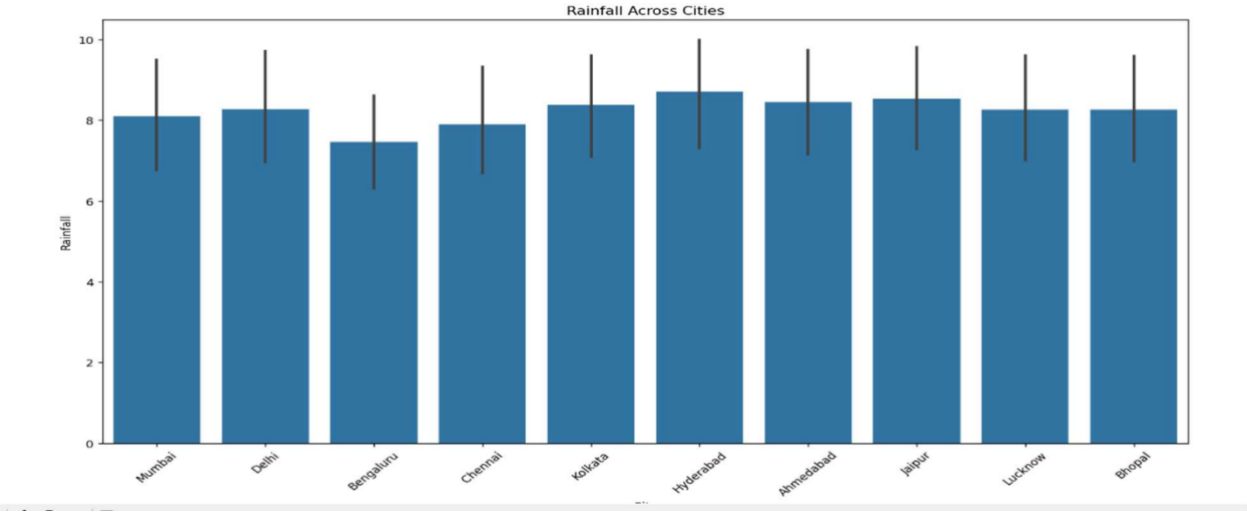


Figure 6: Rainfall comparison

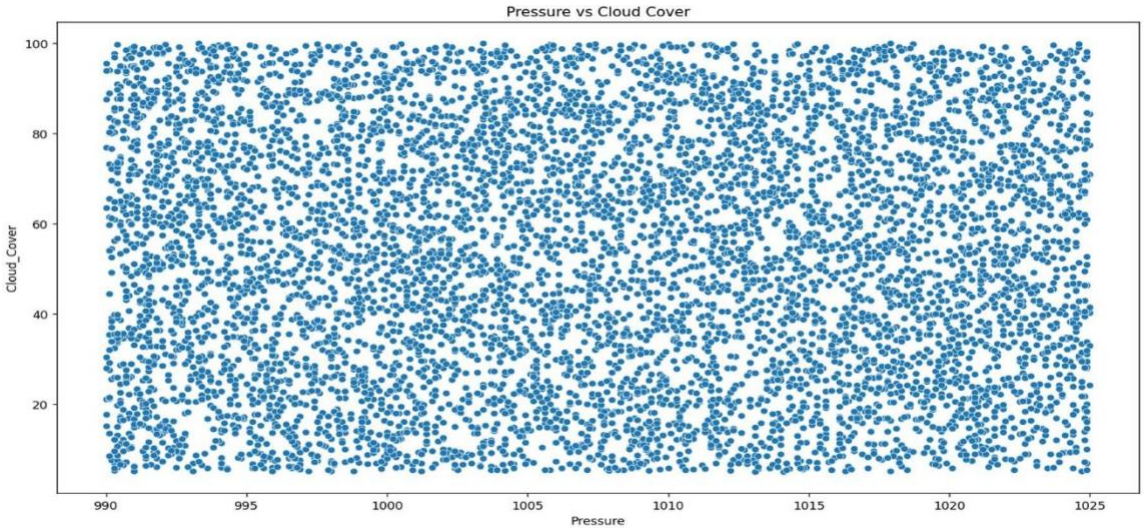


Figure 7: Pressure vs Cloud Cover

8. RESULTS AND FUTURE ENHANCEMENT

The analysis of the weather dataset shows that temperature variables (maximum, minimum, and average) are highly correlated, indicating consistent temperature patterns across cities. Rainfall and AQI exhibit significant variation, suggesting region-specific climatic and environmental conditions. Most other variables, such as humidity, wind speed, and pressure, show weak or no correlation with each other, indicating independent behavior. Scatter plots confirm the absence of strong relationships between variables like humidity and rainfall or wind speed and temperature. Overall, the results highlight that weather patterns are complex, non-linear, and vary significantly across different regions.

8.1. Temperature Analysis

- Maximum temperature reaches up to **45°C**, while minimum goes down to **~10°C**
- Average temperature is around **30°C**
- Strong consistency observed between max, min, and average temperatures

8.2. Rainfall Analysis

- Rainfall varies from **0 mm to ~79.9 mm**
- Many observations show **no rainfall**
- Indicates uneven and region-specific precipitation patterns

8.3. Humidity Analysis

- Humidity ranges from **30% to 95%**
- Wide variation across cities
- No strong link observed with rainfall

8.4. Humidity vs Rainfall Relationship

- Scatter plot shows **no clear pattern**
- High humidity does not always result in rainfall

8.5. AQI Analysis

- AQI ranges from **40 to 349**
- Most values fall in **moderate to poor categories**
- Indicates varying air quality conditions across regions

8.6. Wind Speed vs Temperature

- Wind speed ranges from **2 km/h to 25 km/h**
- No strong relationship with temperature

8.7. Pressure vs Cloud Cover

- No clear trend observed
- Weak or negligible relationship

8.8. Correlation Analysis

- Strong positive correlation between:
  - Temperature_Max, Temperature_Min, Temperature_Avg
- Other variables show **weak or near-zero correlation**

8.9. Overall Observation

- Weather variables behave mostly **independently**
- Patterns are **complex and non-linear**
- Significant variation exists across cities

The project can be improved by applying machine learning models to predict weather conditions and using time series analysis for forecasting. It can also be extended by integrating real-time weather data, using geographical visualizations like maps, and improving data quality with larger datasets. Additionally, developing a user-friendly dashboard can make the analysis more interactive and easier to understand.

Git Hub Link of the project along with the report	https://github.com/mukesh02032007/Weather_Data_Analytics/blob/main/main.py

9. REFERENCES

1. Kaggle Dataset – Indian Climate Dataset 2024–2025. Available at: https://www.kaggle.com/
2. McKinney, W. (2017). *Python for Data Analysis* (2nd ed.). O'Reilly Media.
3. VanderPlas, J. (2016). *Python Data Science Handbook*. O'Reilly Media.
4. Seaborn Documentation. Available at: https://seaborn.pydata.org/
5. Matplotlib Documentation. Available at: https://matplotlib.org/
6. Python Software Foundation. *Python Documentation*. Available at: https://docs.python.org/
7. Pandas Development Team. *Pandas Documentation*. Available at: https://pandas.pydata.org/docs/
8. NumPy Developers. *NumPy Documentation*. Available at: https://numpy.org/doc/
9. Scikit-learn Developers. *Scikit-learn Documentation*. Available at: https://scikit-learn.org/stable/
10. Han, J., Kamber, M., & Pei, J. (2011). *Data Mining: Concepts and Techniques*. Morgan Kaufmann.
11. Géron, A. (2019). *Hands-On Machine Learning with Scikit-Learn & TensorFlow*. O'Reilly Media.
12. Provost, F., & Fawcett, T. (2013). *Data Science for Business*. O'Reilly Media.
13. Aggarwal, C. C. (2015). *Data Mining: The Textbook*. Springer.
14. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An Introduction to Statistical Learning*. Springer.
15. WHO (World Health Organization). *Global Health Observatory Data Repository*. Available at: https://www.who.int/
16. UCI Machine Learning Repository. *Heart Disease Dataset*. Available at: https://archive.ics.uci.edu/
17. IBM. *What is Data Analytics?* Available at: https://www.ibm.com/topics/data-analytics
18. Google Developers. *Data Analytics Learning Resources*. Available at: https://developers.google.com/
19. Microsoft Learn. *Introduction to Data Analysis*. Available at: https://learn.microsoft.com/
20. Towards Data Science. *Articles on Data Analytics and Machine Learning*. Available at: https://towardsdatascience.com/